

Histology of CARDIOVASCULAR SYSTEM (CVS)

CHAPTER OUTLINE

- Introduction
- Histology of heart wall
- Histology of Blood Vessels
 - Arteries
 - Types
 - Histological layers
 - Sensory receptors of arteries
 - Arterioles
 - Capillaries
 - Types, distributions, locations
 - Functions
 - Venules
 - Veins (Types and histological layers)
- Histology of lymphatic vessels and lymphoid organs

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CIRCULATORY SYSTEM

- Ensures continuous uninterrupted movement of the body fluids (blood, lymph & CSF) throughout the body.
- **Functions to:**
 - deliver blood and immune cells throughout the body
 - transport oxygen and nutrients to the body tissues
 - transport carbon dioxide & other metabolic wastes away from the tissues
 - involve in thermoregulation & distribution of molecules such as hormones & cells such as those of the immune system
- **has two functional components:**
 1. Blood Vascular System &
 2. Lymph Vascular System

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The Blood Vascular System/Cardiovascular system

- comprises a circuit of vessels through which flow of blood is maintained by continuous pumping action of the heart.
- includes the heart, arteries, arterioles, capillaries, venules & Veins

The Arterial System

- Constitutes the distribution network of blood to the capillaries beds, which are the main sites of exchange of gases & metabolites between the tissues and blood.
- The cyclical pumping action of the heart produces a pulsatile blood flow in the arterial system.
- During contraction of ventricles (systole), the arterial wall is expanded.

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The Arterial System

- Between the subsequent ventricular contractions (diastole), the arterial wall recoils back.
- This expansion and recoil is a function of elastic tissue within the wall of the arteries.**
- The flow of blood to various organs and tissues may be regulated by varying the diameter of distributing vessels.
- This function is performed by the circumferentially disposed smooth muscle of vessel wall and is principally under the control of the sympathetic NS and adrenal medullary hormones.

The Venous System

- Returns blood from the capillaries to the heart

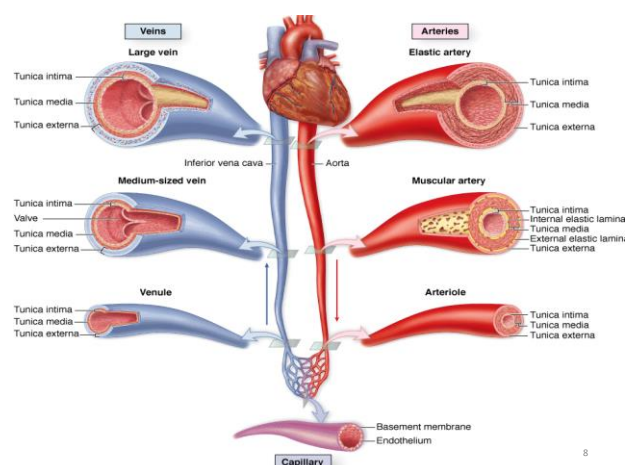
THE LYMPHATIC VASCULAR SYSTEM

- Is merely a passive drainage system for returning excess extravascular fluid, **the lymph**, to the blood vascular system.
- The lymph contains lymphocytes and other immunologic factors and proteins.
- Unlike the blood, lymph is circulated in one direction, towards the heart only.
- Unlike the blood vascular system, it has no intrinsic pumping mechanism.

General Histology of Circulatory system

- The whole circulatory system has a common basic structure:
 - Anatomically:** outer **wall** & the interior **lumen**, a central fluid containing hollow space surrounded by the wall.
 - Histologically:** **three basic layers or tunics** (L. *tunica*, coat) in the wall:

1. Tunica intima (interna)
2. Tunica media (middle wall)
3. Tunica adventitia (externa)



1. Tunica Intima

- is the innermost layer, very close to the lumen
- consists of an **endothelium** (lining the interior), **subendothelial CT** layer and an **internal elastic lamina**.

a) Endothelium:

- a single layer of extremely flattened endothelial cells (simple squamous epithelium) lining the lumen.
- supported by a basal lamina & a delicate fibroelastic CT
- the endothelial cells are polygonal and elongated in the direction of blood flow,
- the nucleus causes that part of the cell to bulge into the lumen

b) Subendothelial CT layer (*subendothelium*):

- is a thin loose areolar CT layer beneath the endothelium, with occasional smooth muscle cells formerly known as **myointimal cells**.

c) Internal Elastic Lamina- thin elastic tissue layer

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2. Tunica Media (middle wall)

- An intermediate muscular or elastic tissue wall
- Consists of circumferentially or spirally arranged **smooth muscle fibers**, **collagen type III (reticular)** & **elastic fibers**.
- In arteries, the tunica media is separated from the tunica intima by the **internal elastic lamina/membrane**.
- In larger vessels, tunica media is separated from the tunica adventitia by an **external elastic lamina**.
- This layer exhibits the greatest variation from one part of the system to another.
 - For example it is totally absent in capillaries and postcapillary and collecting venules, but comprises almost the whole mass of the heart and large arteries.
- Blood flow is predominantly influenced by variation in activity of this muscular layer.

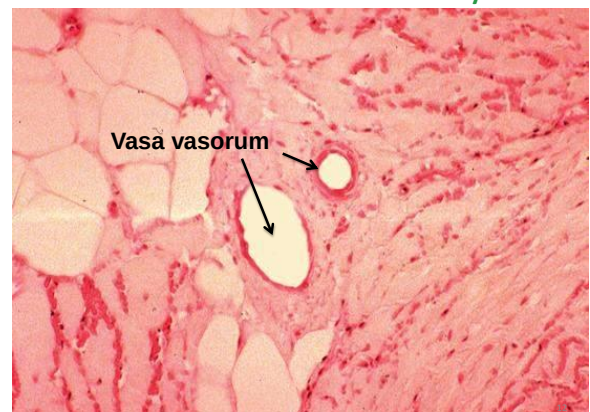
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3. Tunica Adventitia (Externa)

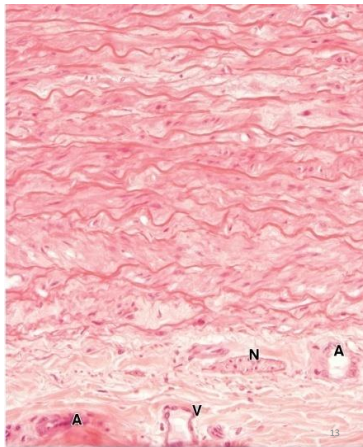
- the outer most supportive tissue layer of vessels
- principally made up of fibroelastic CT with collagenous and elastic fibers which blend with the surrounding supportive tissue.
- In larger vessels, it contains the **vasa vasorum and nervi vasorum**.
 - The tissues of the walls of large vessels cannot be sustained by diffusion of nutrients from their lumina and are thus supplied by small arteries called vasa vasorum (i.e. vessels of vessels).
- **The vasa vasorum**
 - Derived either from the main vessel itself or from adjacent arteries
 - Gives rise to a capillary network within the tunica adventitia which may extend into the tunica media.
- Also contains nonmyelinated vasomotor nerves (called **nervi vasorum/vascularis**) which regulate the blood pressure.

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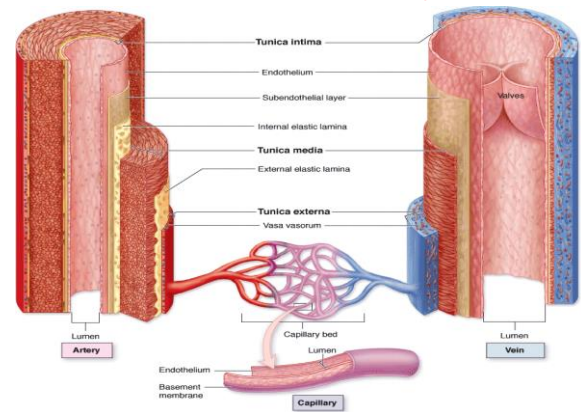
Tunica adventitia of Elastic Artery



Vasa vasorum: walls of the larger vessels, as the aorta, contain in the tunica adventitia a supply of microvasculature to bring O_2 and nutrients to local cells too far from the lumen to be nourished by blood there. These arterioles (A), capillaries and venules (V) constitute the vasa vasorum (vessels of vessels). The adventitia of large arteries is also supplied more sparsely with small sympathetic nerves (N) for control of vasoconstriction. X100.



Walls of arteries, veins, and capillaries

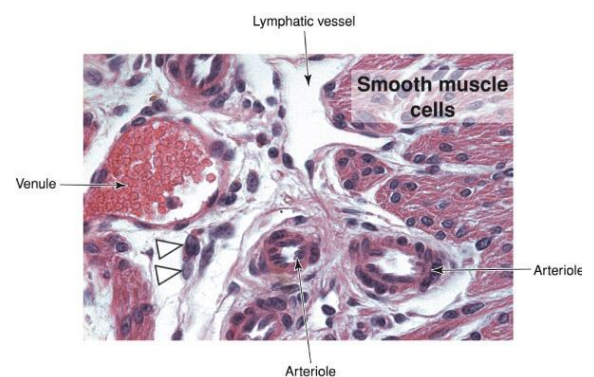


1. Macrovasculature:

- vessels that are larger than 0.1 mm in diameter
- Includes elastic arteries, muscular arteries, small arteries, veins and lymphatics.

1. Microvasculature

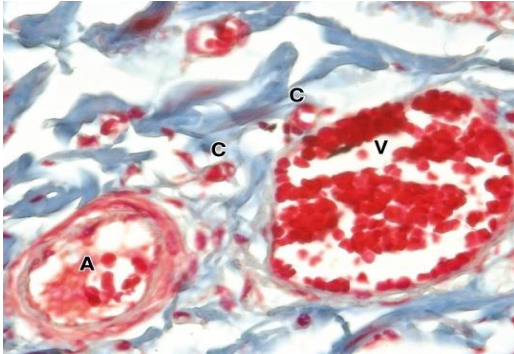
- Less than 0.1 mm in diameter
- Includes arterioles, metarterioles, capillaries, and venules (visible only with a microscope).
- particularly important functionally, being the site of exchanges between blood and the surrounding tissues both under normal conditions and during inflammatory processes.



Small blood vessels as components of microvasculature (arterioles and venules) surrounded by components of loose connective tissue. The arrowheads point to fibroblasts. H&E stain. Low magnification.

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Vessels of the microvasculature: Arterioles (A), small capillaries (C) and venules (V) make up the microvasculature where, in almost every organ, exchange takes place between blood and the interstitial fluid of the tissues. X200. Trichrome.

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HEART

- A hollow muscular organ that contracts rhythmically.
- Central organ of the circulatory system.
- Contains hollow spaces (chambers) surrounded by wall
- Contains its own conduction system.
- Has 3 basic layers in its wall:
 1. Inner endocardium,
 2. Middle myocardium and
 3. Outer Epicardium.

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1. Endocardium:

- homologues with the intima of the blood and lymphatic vessels
- consists of a single layer of squamous endothelial cells (endothelium) resting on a thin layer of loose CT (subendothelial layer) containing elastic and collagen fibers as well as some smooth muscle cells.
- Connecting this subendothelial layer to the myocardium is additional connective tissue layer (often called the **subendocardial layer**) containing veins, nerves, and branches of the impulse-conducting system of the heart.

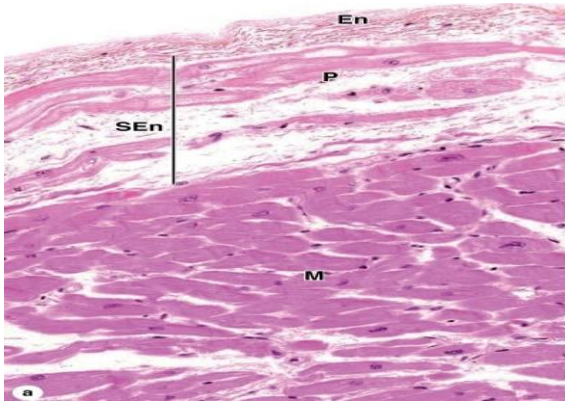
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2. Myocardium:

- is the thickest of the tunics and by far greater than the other two layers &
- contains contractile cardiac muscle cells and impulse generating-conducting cells.
- is much thicker in the ventricles than in the atria.
- It is thickest in the left ventricle
- consists of cardiac muscle tissue cells arranged in layers that surround the heart chambers that are interwoven in layers to form a complex spiral pattern.
- the arrangement of these muscle cells is extremely varied, so that in sections cells are seen to be oriented in many directions.

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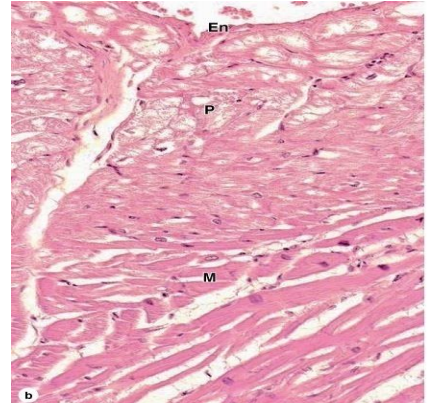
Endocardium (En) and Myocardium (M)



Subendocardial (SEn) conducting network & Purkinje fibers (P) ²¹

Purkinje fibers

intermingling with contractile fibers within the myocardium (M). Along with the nodes of specialized cardiac muscle in the right atrium which generate the electrical impulse, the network of conducting fibers comprises the conducting system of the heart. X200. H&E.



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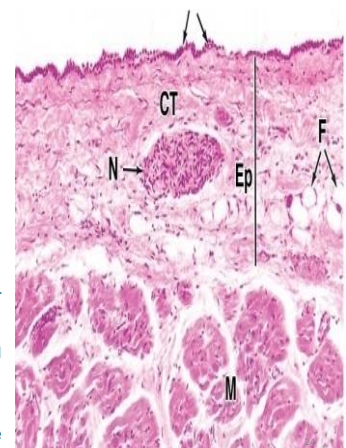
3. Epicardium

- is the serous covering of the heart.
- Externally, it is covered by modified simple squamous epithelium (mesothelium) supported by a thin layer of loose CT (submesothelial layer), which contains veins, nerves, and many adipocytes.
- Generally surrounded by adipose tissue.
- corresponds to the visceral layer of the serous **pericardium**, the serous membrane in which the heart lies.
- In the space between the pericardium's visceral layer (epicardium) and its parietal layer is a small amount of lubricant serous fluid that facilitates smooth movement of the heart.

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Epicardium or visceral

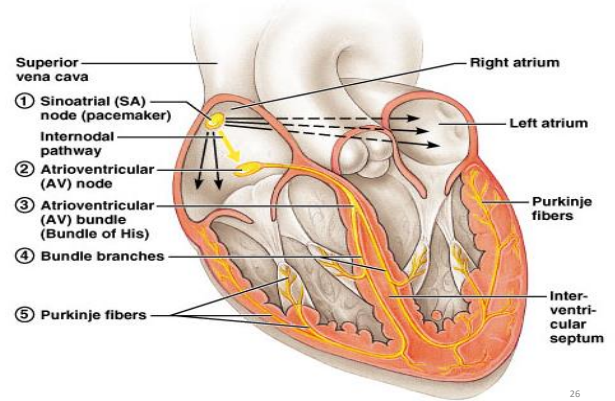
pericardium: The external tunic of the heart, the epicardium, is the site of the coronary vessels and contains considerable adipose tissue. This section of atrium shows part of the myocardium (M) and epicardium (Ep). The epicardium consists of loose connective tissue (CT) containing both autonomic nerves (N) and fat (F). The epicardium is the visceral layer of the serous pericardium and is covered by the simple squamous-to-cuboidal epithelium (mesothelium) (arrows) that also lines the pericardial space. These mesothelial cells secrete a lubricate fluid that prevents friction as the beating heart contacts the parietal pericardium on the other side of the pericardial cavity.



Histological Layers of Atrial Wall: En=endocardium; M= myocardium which is very thin ; Ep= epicardium



Heart Conducting System

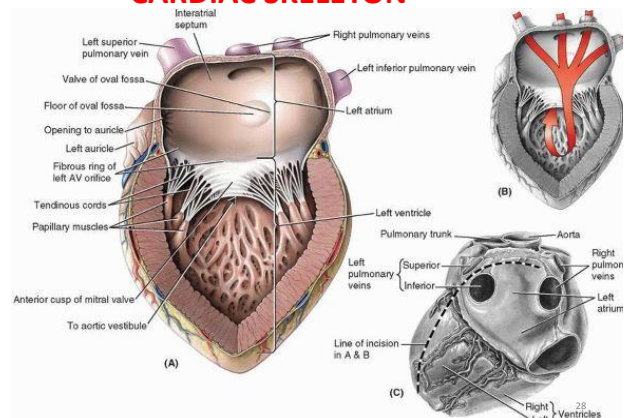


CARDIAC SKELETON

- a fibro-elastic collagenous CT skeleton that serves as support and site of origin & attachment for the cardiac muscles and heart valves.
- its principle parts are:
 - **Septum membranaceum** (membranous portions of interatrial, AV and interventricular septa)
 - **Two Trigona fibrosa** (right and left fibrous trigona)
 - **Four Annuli fibrosa** (4 fibrous rings) - one ring around each of the four cardiac valves
- The cardiac valves are made up of two layers of endothelial cells sandwiching a central portion of dense fibrous connective tissue core, known as **Lamina Fibrosa**.

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CARDIAC SKELETON



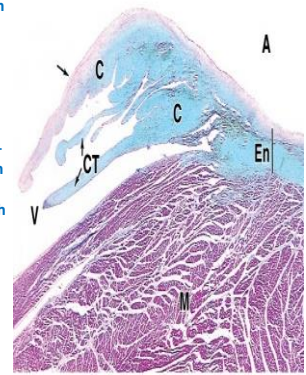
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- The cardiac valves consist of a central core of dense fibrous connective tissue (containing both collagen and elastic fibers) [known as **lamina fibrosa**] lined on both sides by endothelium.
- The bases of the valves are attached to strong fibrous rings that are part of the **fibrous skeleton**.
- This dense, fibrous region around the heart valves anchors the base of the valves and is the site of origin and insertion of the cardiac muscle fibers

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Valve leaflet and fibrous skeleton:

The fibrous skeleton of the heart consists of masses of dense CT in the endocardium which anchors the valves and surrounds the two atrioventricular canals, maintaining their proper shape. Section through a leaflet of the left AV valve (arrows) shows that valves are largely dense connective tissue (C) covered with a thin layer of endothelium. The collagen-rich CT of the valves is stained pale green here and is continuous with the fibrous ring of CT at the base of the valves, which fills the endocardium (En) of this area between the atrium (A) and ventricle (V). The chordae tendinae (CT), small strands of CT which bind distal parts of valve leaflets are also seen here. The interwoven nature of the cardiac muscle fibers, with many small fascicles, in the myocardium (M) is also shown. X20. Masson trichrome.



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BLOOD AND LYMPHATIC VESSELS

- Have a central Lumen surrounded by a wall.
- Lumen – central blood-containing space surrounded by the tunics
- Arteries, arterioles, muscular venules, veins and lymphatics are composed of three histological layers (tunics) in their wall:
 1. **Tunica interna (intima)** (Internal layer, closes to the lumen)
 2. **Tunica media** (Middle layer)
 3. **Tunica externa (Adventitia)**- Outermost layer

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Histological Layers of Vascular Wall

1. TUNICA INTIMA (INTERNA)

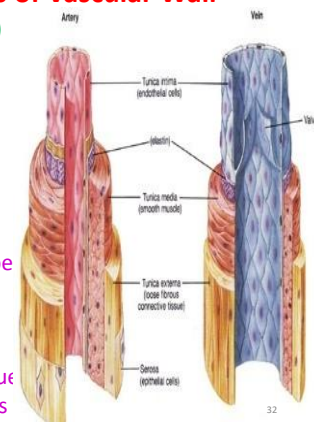
- Lined by simple squamous epithelium (**endothelium**) resting on basement membrane
- subendothelial loose CT
- internal elastic lamina

2. TUNICA MEDIA

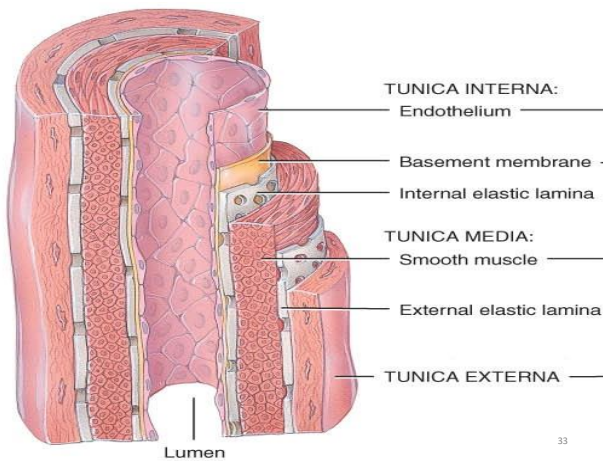
- smooth muscle, reticular fibers & external elastic lamina

3. TUNICA ADVENTITIA/ EXTERNA

- Fibro-elastic supportive tissue with elastic & collagen fibers



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Tissue Components of the Vascular Wall

- The vascular wall is composed of three basic structural constituents: **epithelium (endothelium)**, the **muscle tissue**, and **connective tissue**, which includes both collagenous and elastic elements.
- The amount and arrangement of these tissues within the blood vascular system are influenced by **mechanical factors**, represented primarily by **blood pressure**, and **metabolic factors**, which reflect the local needs of the tissues.
- These tissues are all present in different proportions in the vascular wall, except for **capillaries**, **postcapillary** and **collecting venules**, in which the only structural elements represented are the endothelium, its basal lamina, and **pericytes**.

THE ENDOTHELIUM:

- is a special type of epithelium that acts as a semipermeable barrier between two internal compartments: the blood plasma/lymph and the interstitial tissue fluid.
- is highly differentiated to mediate and actively monitor the bidirectional exchange of small molecules and restrict the transport of some macromolecules.
- Besides their role in the exchanges between blood and tissues, the endothelial cells perform several other functions including:
 1. **production and release of vasoactive factors that affect the vascular tone**, such as nitric oxide, endothelins, and vasoconstrictive agents,
 2. **conversion of circulating angiotensin I to angiotensin II**
 3. **Growth factors** such as **vascular endothelial growth factor (VEGF)** help maintain the vasculature, regulate the formation of the vascular system from embryonic **mesenchyme (vasculogenesis)** and promote capillary outgrowth from existing vessels (**angiogenesis**) under normal and pathologic conditions in adults.

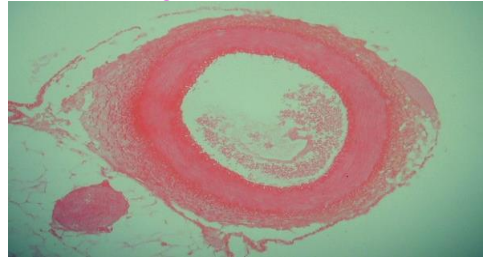
- The endothelium also has an antithrombogenic action, preventing blood coagulation.
- Although morphologically similar, the endothelial cells of different blood vessels exert their various functional properties differently.
- Endothelial cells, especially those of arteries, contain very small, elongated **vesicles** called **Weibel-Palade bodies**, which contain **selectin** and **von Willebrand factor** involved in blood coagulation.
- When endothelial cells are damaged by atherosclerotic lesions, for example, the uncovered subendothelial CT induces the aggregation of blood platelets.
- This aggregation initiates a cascade of events that produce fibrin from circulating fibrinogen.

- An intravascular clot, or **thrombus** (plural, thrombi), is formed that may grow until there is complete obstruction of the local vascular flow.
- From this thrombus, solid masses called **emboli** (singular, embolus) may detach and be carried by the blood to obstruct distant vessels.
- In both cases the vascular flow may stop, a potentially life-threatening condition.
- Thus, the integrity of the endothelial layer preventing the contact between platelets and the subendothelial connective tissue is an important antithrombogenic mechanism.

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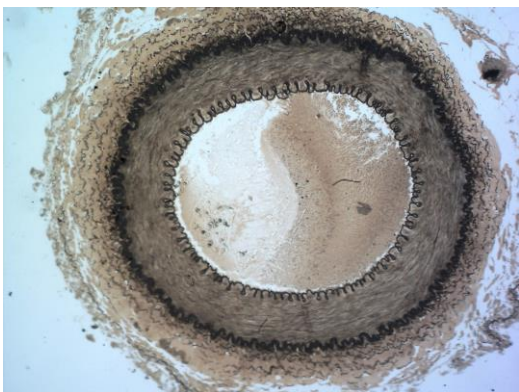
ARTERIES

- Thick walled vessels, the wall thickness approximates the lumen diameter.
- carry blood away from the heart
- Generally delivers oxygenated blood with the exception of the pulmonary and umbilical arteries.
- Classified according to their size



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TS of Artery



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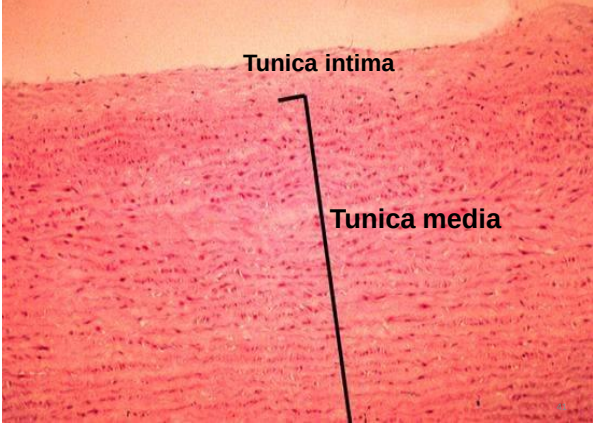
TYPES OF ARTERIES

1. ELASTIC (CONDUCTING) ARTERIES:

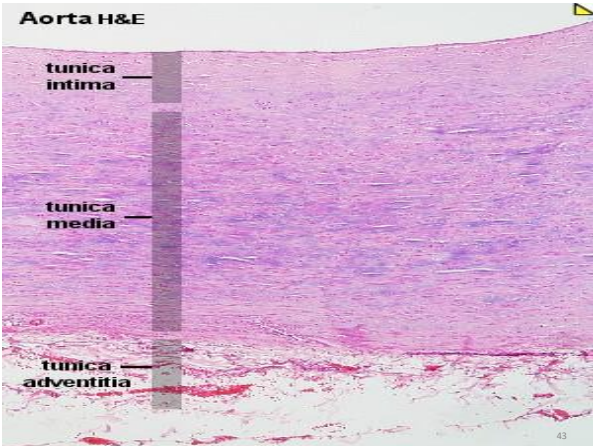
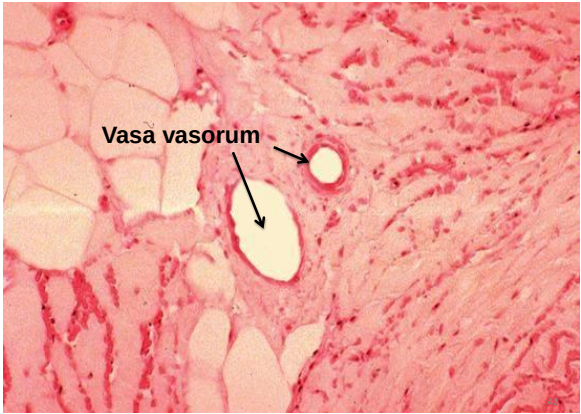
- are the largest arteries(>10 mm) and include the aorta and its large branches, and Pulmonary arteries.
- have a yellowish color in fresh preparations owing to the accumulation of elastin in their media.
- have the following characteristics:
 1. The intima is thick and endothelium contains microvilli.
 2. The subendothelial CT is thick and are placed longitudinally oriented to cope with the pressure by permitting elongation of the vessel.
 3. An internal elastic lamina, although present, not easily discerned.
 4. Media contains multiple layers of concentrically arranged smooth muscles separated by fenestrated elastic membranes that increase with ones age and blood pressure (absent at birth, 40 to 70 lamellae in the adult).
 5. External limiting membrane/elastic lamina is totally absent.

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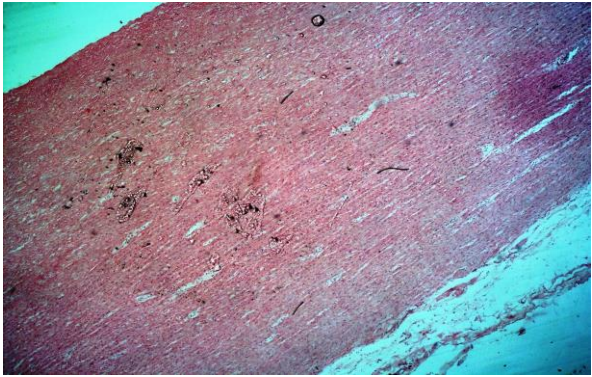
Tunica intima and tunica media of Elastic Artery



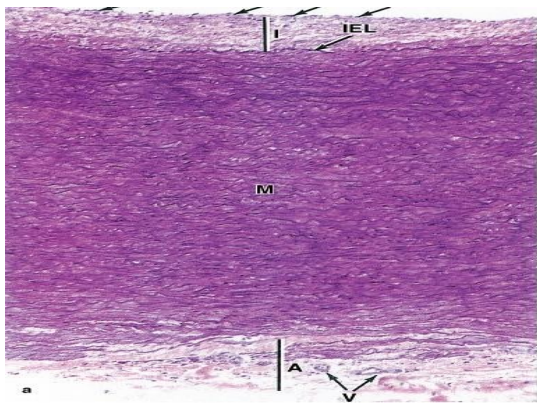
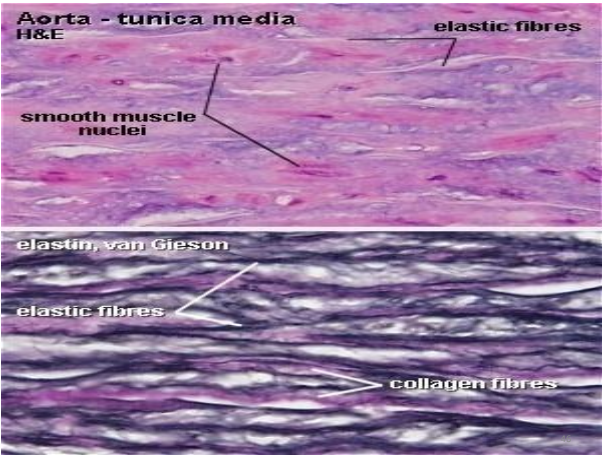
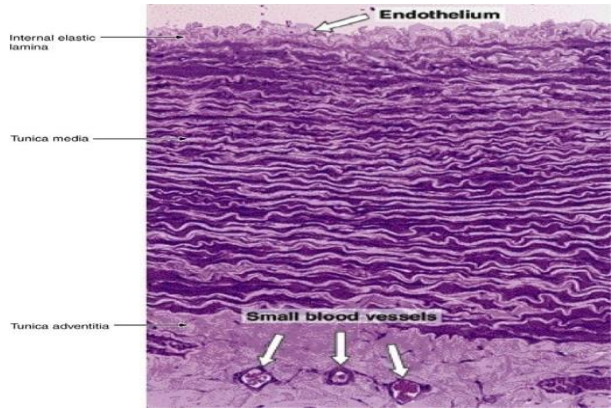
Tunica adventitia of Elastic Artery



Histological Layers of Aorta

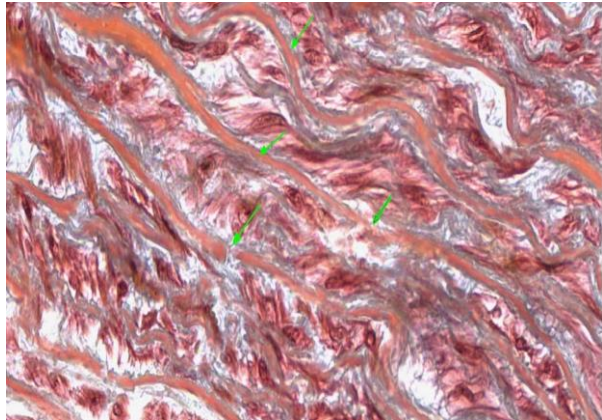


Elastic C T in the wall of large artery (Aorta)



Comparison of the three major layers or tunics in the largest artery

Fenestrated Elastic laminae in the wall of Aorta, Mallory Trichrome stain



2. MEDIUM-SIZED (MUSCULAR OR DISTRIBUTING) ARTERIES

The best examples are the femoral artery and brachial artery.

1- Tunica intima:

- Endothelium
- Thin layer of subendothelium, containing collagen, elastic and smooth muscle fibers.
- There is a prominent bifid internal elastic lamina.

2- Tunica media:

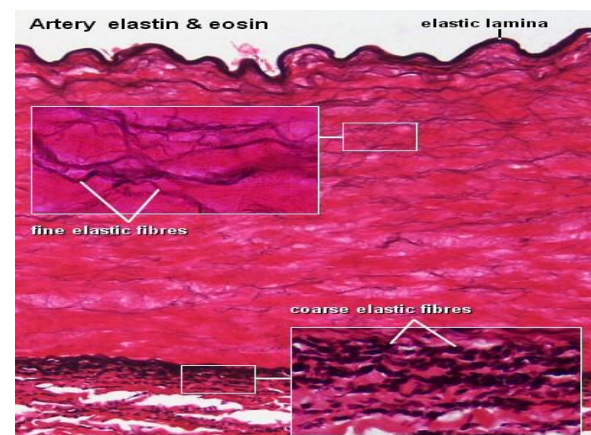
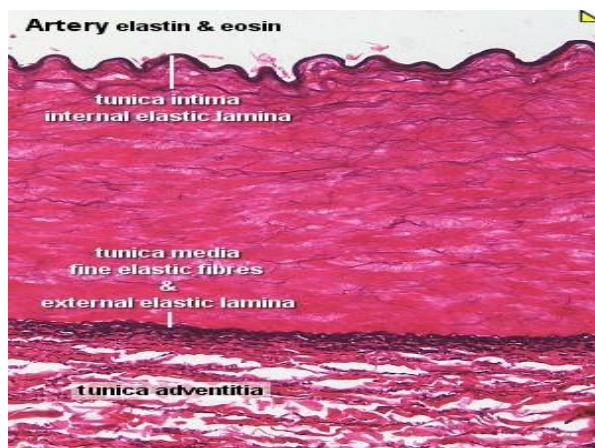
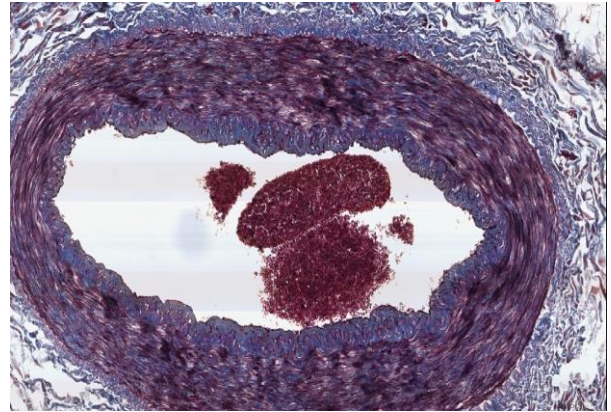
- It is thick layer of concentrically arranged smooth muscle fibers.
- Elastic and reticular fibers are located in between the muscle fibers
- In large medium-sized arteries, there is an external elastic lamina between the media and the adventitia.

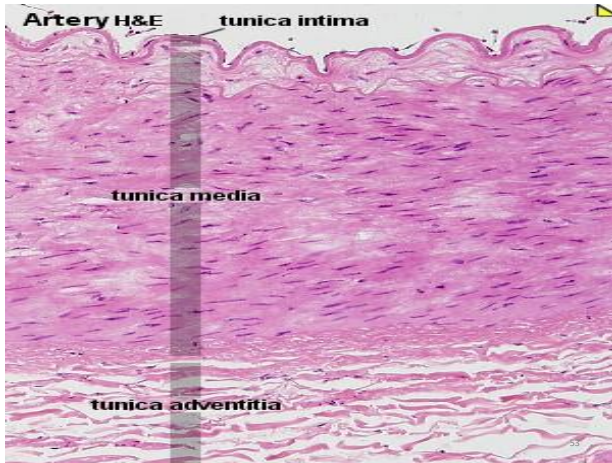
3- Tunica adventitia:

- It consists of collagen, elastic fibers, few fibroblasts, fibrocytes, lymphatics, small vessels and nerves.

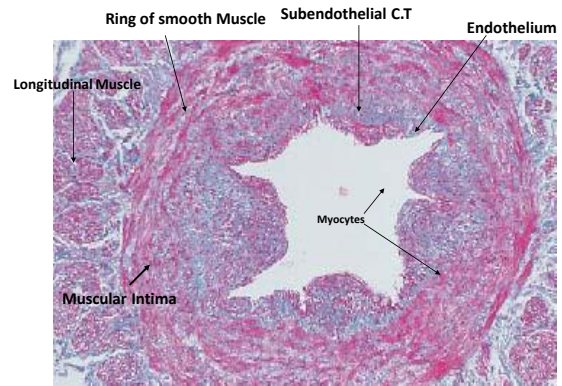
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TS of a Muscular Artery





Arterial Constrictor



3. Small Arteries

- are small aa having an outer diameter ranging 0.1 – 2 mm
- generally thin walled and narrow lumen; the lumen diameter approximates the thickness of the wall.

1. The tunica intima:

- has endothelium, subendothelial CT, smooth muscle and an internal elastic limiting membrane (internal elastic lamina).

2. The tunica media:

- consists of concentrically disposed, as many as 8 layers of smooth muscle fibers and collagen fibers, no EEL.

3. The tunica adventitia:

- thin layer of loose collagenous CT which merges imperceptibly with neighboring connective tissue.

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4. Arterioles

- are small unnamed branches of artery having an outer diameter less than 0.5 mm (10-500 μ m)
- generally thin walled and narrow lumen; the lumen diameter approximates the thickness of the wall.
- are the site for blood pressure regulation

1. The tunica intima:

- Consists of endothelium, basal lamina with extremely thin proteoglycans, lack an internal elastic limiting membrane (no IEL).

2. The tunica media:

- consists of concentrically disposed one to three layers of smooth muscle fibers and no external elastic lamina.

3. The tunica adventitia:

- thin ill defined layer of loose collagenous CT sheath

3. Arterioles

- are small unnamed branches of artery having an outer diameter less than 0.5 mm
- generally thin walled and narrow lumen; the lumen diameter approximates the thickness of the wall.
- are the site for blood pressure regulation

1. The tunica intima:

- has no subendothelial layer and lack an internal elastic limiting membrane (no internal elastic lamina).

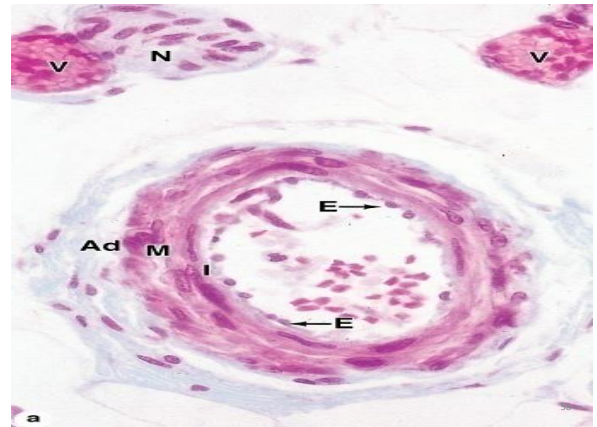
2. The tunica media:

- consists of concentrically disposed one or two layers of smooth muscle fibers and no external elastic lamina.

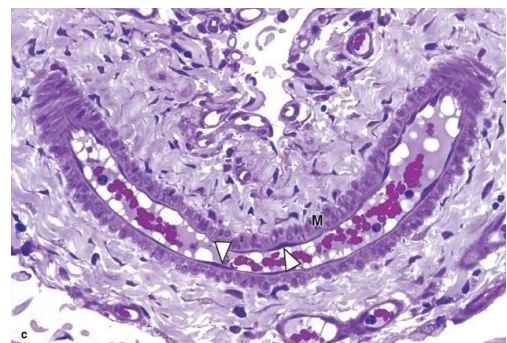
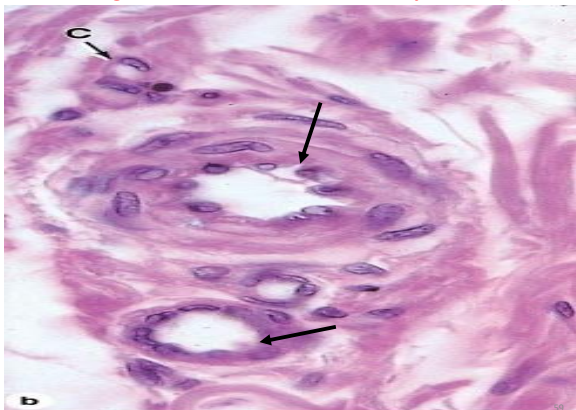
3. The tunica adventitia:

- thin layer of loose collagenous CT which merges imperceptibly with neighboring connective tissue.

Arteriole, venules (v) and a small nerve (N)

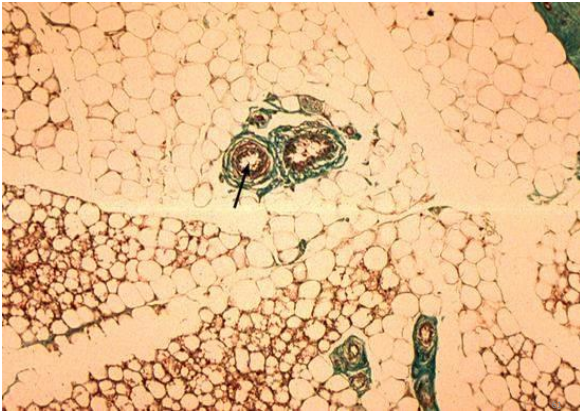


Large and small Arterioles, and capillaries (C)

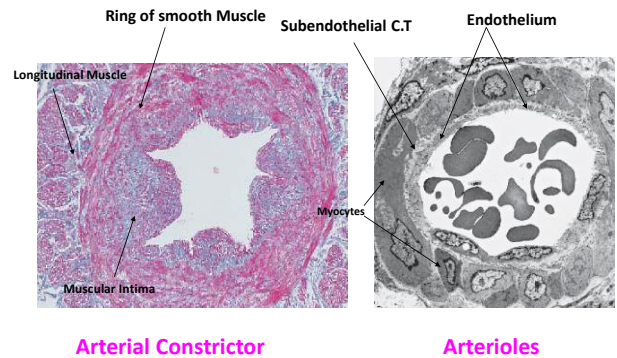


A large mesenteric arteriole is cut obliquely and longitudinally and clearly shows the endothelial cells (arrow heads) and one or two layers of smooth muscle cells (M) cut transversely. Adventitia merges imperceptibly with neighboring connective tissue.

ARTERIOLE IN ADIPOSE TISSUE



Arteries



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ARTERIAL SENSORY RECEPTORS

Carotid Sinuses:

- are slight dilatations of the internal carotid arteries which contain **baroreceptors** detecting increases in blood pressure.
- the tunica media of each carotid sinus is thinner, allowing greater distension when blood pressure rises, and the intima and adventitia are rich in sensory nerve endings from **cranial nerve IX, the glossopharyngeal nerve**.
- The afferent nerve impulses are processed in the brain to trigger adjustments in vasoconstriction that return pressure to normal.
- Similar baroreceptors occur in the arch of aorta and other large arteries.

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The Carotid Bodies (Glomus Bodies):

- are small, ganglia-like structures (paraganglia) near the bifurcation of **the common carotid arteries** that contain **chemoreceptors** sensitive to blood CO_2 & O_2 concentrations.
- a network of sinusoidal capillaries is intermixed with **glomus (type I) cells** containing numerous dense-core vesicles with **dopamine, serotonin, and adrenaline**.
- Dendritic fibers of cranial nerve IX, the glossopharyngeal nerve, synapse with the glomus cells.
- The sensory nerve is activated by neurotransmitter release from glomus cells in response to changes in the sinusoidal blood: increased CO_2 , decreased O_2 , or increased H^+ levels.
- **Aortic bodies** located in the wall of aortic arch are similar in structure and function to carotid bodies.

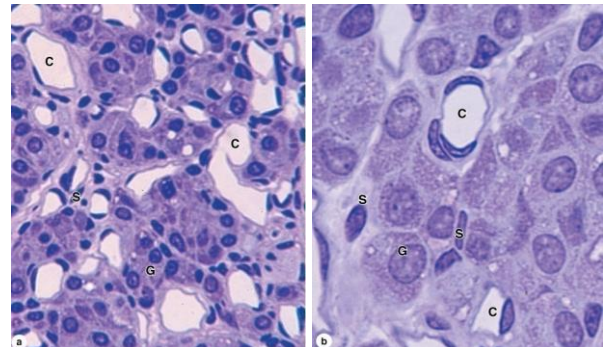
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Glomus body (The next slides) (a) and (b): Specialized regions in the walls of specific arteries contain cells that act as chemoreceptors providing information to the brain regarding blood chemistry.

The Glomus Bodies

- are two small (0.5–5 mm diameter) ganglion-like structures found near the bifurcations of the common carotid arteries.
- contain many large sinusoidal capillaries (C) intermingled with clusters of large glomus cells (G) with round nuclei and cytoplasm filled with vesicles of various neurotransmitters that are clearly seen in (b).
- **Supportive sheath cells (S)** with elongated nuclei are associated with the groups of glomus cells.
- Glomus cells form synaptic connections with dendritic fibers of the glossopharyngeal nerve.
- Changes in the CO_2 , O_2 and H^+ conc. in the sinusoidal blood are detected by the glomus cells, which then release neurotransmitter that activates the sensory nerve to relay this information to the brain.

65



Glomus body: Specialized region in the walls of specific arteries contain cells that act of chemoreceptors providing information to the brain regarding blood chemistry.

66

Arterial Degenerative Alterations

1. Atherosclerotic lesions

- are characterized by focal thickening of the intima, proliferation of smooth muscle cells and connective tissue elements.
2. Certain arteries irrigate only specific areas of specific organs and obstruction of this blood supply results in **necrosis** (death of tissues from a lack of metabolites).
- These **infarcts** occur commonly in the heart, kidneys, cerebrum, and certain other organs.
3. When the tunica media of an artery is weakened by an embryonic defect, disease, or lesion, the wall of the artery may dilate extensively.
- As this process of dilatation progresses, it becomes an **aneurysm**. Rupture of the aneurysm brings severe consequences and may cause death.

67

CAPILLARIES

- a single layer of endothelial cells rolled up in the form of a tube to form endothelium of blood or lymph capillaries.
- Blood capillaries **interconnect** an arteriole to a venule.
- the average diameter of capillaries varies from 4 to 40 μm (true capillaries= 4 -10 μm in diameter).
- Altogether blood capillaries comprise over 90% of all blood vessels in the body, with a total length of nearly 96,000 km (60,000 miles).
- The total diameter of all capillaries is approximately 800 times larger than that of the aorta.
- The velocity of blood in the aorta averages 320 mm/s, but in capillaries blood flows only about 0.3 mm/s.

68

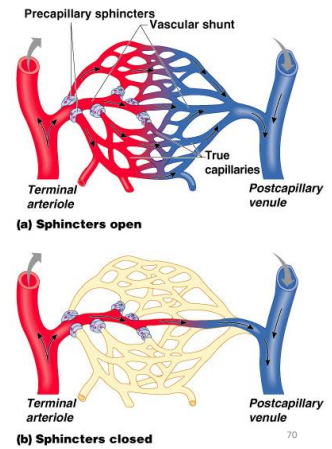
BLOOD CAPILLARIES

- they anastomose freely and form rich networks.
- contain most of the blood
- the endothelial cells usually rest on a **basal lamina**.
- the endothelial cells are polygonal and elongated in the direction of blood flow and the nucleus usually bulges to the lumen.
- the EM of the endothelial cells is characterized by free ribosomes, small Golgi, few mitochondria, RER, intermediate filaments and microfilaments.
- The cells are held together by fascia occludens & gap junctions.

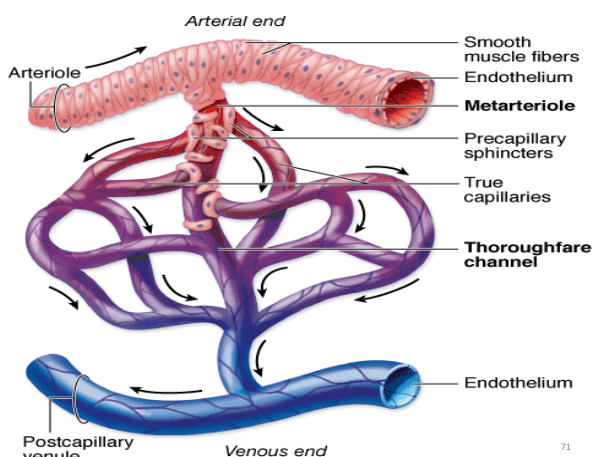
69

Capillaries

- Composed of cells (endothelium) and basement membrane.
- No tunica media or tunica adventitia.
- A capillary bed is a network of 10-100 capillaries.
- **Precapillary sphincters** control the flow of blood through the capillary bed

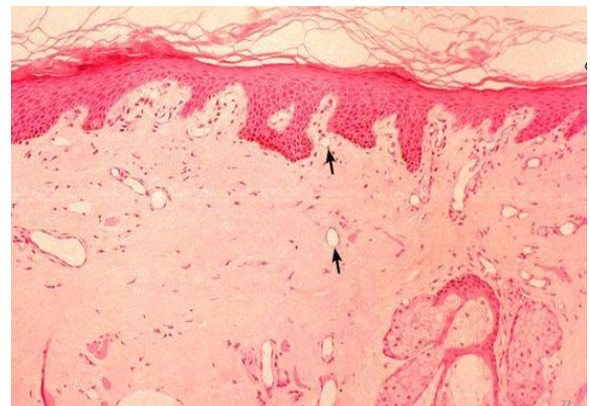


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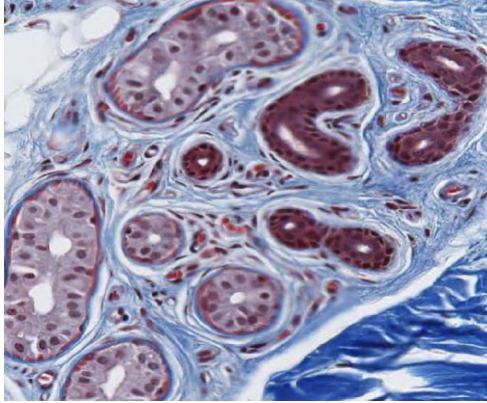


71

Blood capillaries (arrows)



Blood capillaries



73

TYPES OF CAPILLARIES

- Different tissues contain different types of capillaries based on their nutritional demand.
- Three types of blood capillaries depending of structure of the endothelial cells and associated basal lamina:

1. Continuous/Somatic capillaries
2. Fenestrated/perforated-Visceral capillaries
3. Sinusoidal capillaries

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Types of Blood Capillaries

1. Continuous (somatic) capillaries:

- The most common type varying from 5 to 10 μm in diam.

Characterized by

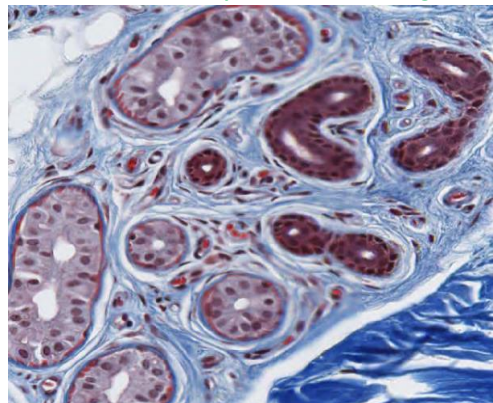
- Continuous endothelium and tight junctions (Fascia occludens) between the endothelial cells with overlapping folds between them.
- the absence of fenestrae in the endothelial cells (lack pores in the wall of capillaries)
- Complete basal lamina
- Many transcytotic pinocytotic vesicles on both the luminal and the basal surfaces of the endothelial cells.

Locations:

1. Muscular tissues
2. Connective tissues
3. Cortex of thymus
4. Testes
5. Exocrine glands
6. CNS (here, the endothelial cells are free of pinocytic vesicles)
7. Adipose tissue

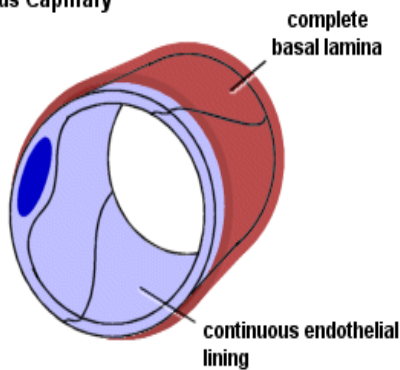
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Continuous blood capillaries in Exocrine glands

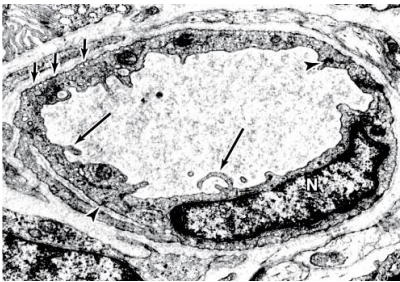


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Continuous Capillary

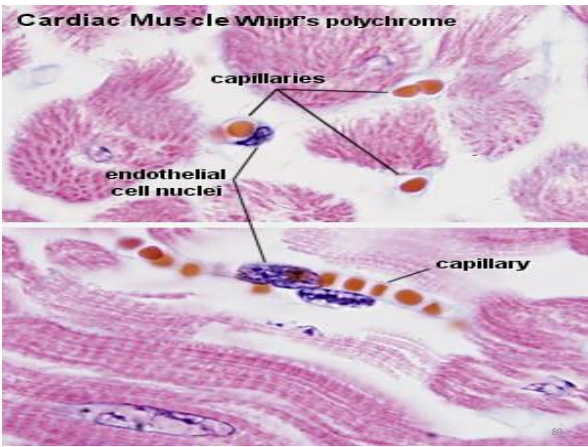
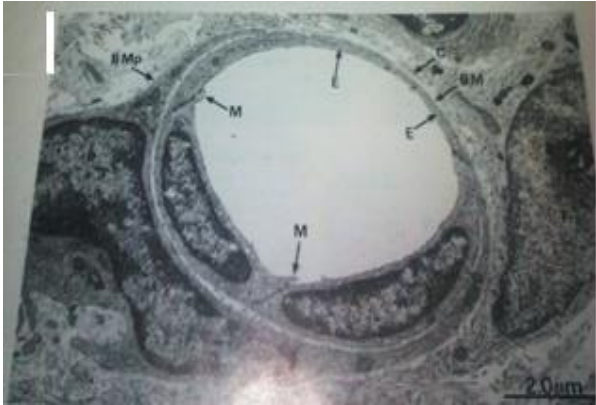


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Continuous capillaries exert the tightest control over what molecules leave across their walls. TEM shows a continuous capillary in transverse section. A nucleus (N) is prominent, but tight or occluding junctions along overlapping folds between two cells can also be seen (arrowheads). Numerous transcytotic vesicles are evident (small arrows). The long arrows show extensions of broad cytoplasmic sheets suggesting phagocytosis, which is consistent with the presence of vacuoles and electron-dense lysosomes. All material that crosses continuous capillary endothelium must pass *through* the cells, usually by diffusion or transcytosis. ⁷⁵

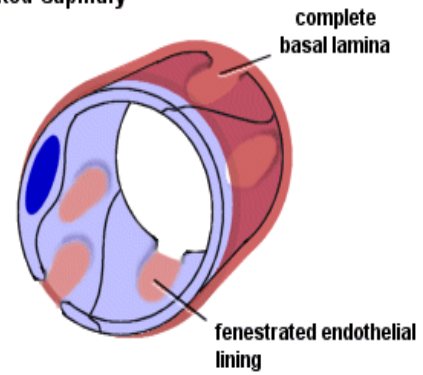
Continuous Capillary



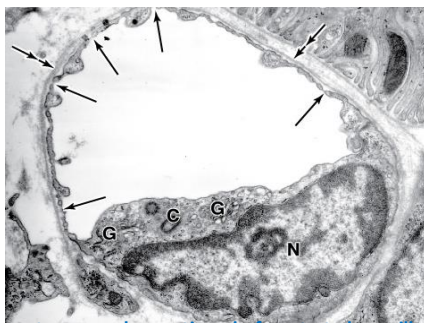
2. Fenestrated/Visceral Capillaries

- are perforated capillaries found in tissues that have rapid exchange of substances with blood.
 - allow a freer exchange of molecules than continuous capillaries.
 - are specialized for uptake of molecules such as hormones in endocrine glands or for outflow of molecules such as in the kidney's filtration system.
- 1- The capillaries are characterized by the presence of large fenestrae (about 60-80 nm in diameter) in endothelial cells.
 - 2- There is a continuous basal lamina
 - 3- The fenestrae may be closed by diaphragm or not.
 - 4- They are present in the:
 - A- Kidney (glomerular capillaries)-not closed by diaphragm
 - B- Intestines, pancreas, gall bladder
 - C- Endocrine glands, synovial membranes, choroid plexus

Fenestrated Capillary



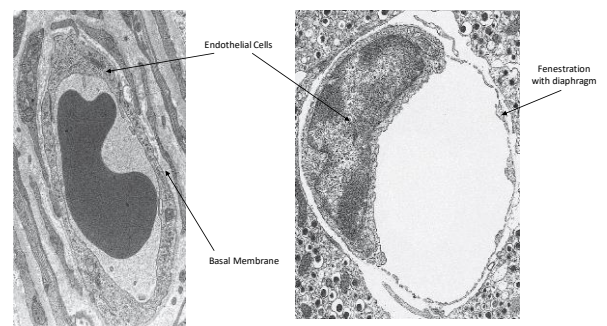
82



TEM of a transversely sectioned fenestrated capillary in the peritubular region of the kidney shows many typical fenestrae closed by diaphragms (arrows), with a continuous external lamina on the outer surface of the endothelial cell (double arrows). In this endothelial cell the Golgi apparatus (G), nucleus (N), and centrioles (C) can be seen.

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Capillaries



Continuous Capillary

Fenestrated Capillary

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3. Sinusoidal or Discontinuous Capillaries

- permits maximal exchange of macromolecules as well as cells between tissues and blood.
- have the following characteristics:
 - very thin endothelium and the endothelial cells nuclei are more difficult to find than in smaller capillaries.
 - have large fenestrae without diaphragms;
 - the cells form a discontinuous layer and are separated from one another by wide spaces;
 - the basal lamina is also discontinuous.
- Sinusoids are irregularly shaped (a tortuous course) and have diameters as large as 30–40 μm , much greater than those of other capillaries, properties which further slow blood flow at this site.
- found in the liver, spleen, placenta, some endocrine glands, and bone marrow & other lymphoid organs.

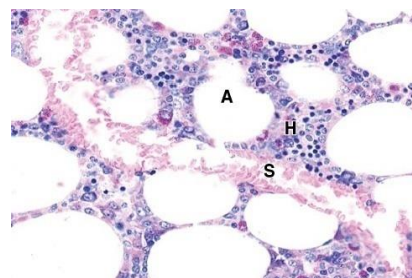
85

Sinusoidal Capillaries

1. They are tortuous dilated capillaries (about 30–40 μm in diameter) which lead to slowing of the blood circulation.
2. They are also known as sinusoids.
3. Present in the placenta, liver, some endocrine glands and the lymphoid organs (spleen, bone marrow, lymph nodes, tonsils).
4. The endothelial wall shows multiple fenestrae without diaphragm.
 - The liver sinusoids reveal abundant grouped fenestrations.
5. The basal lamina is discontinuous and large discontinuities between the cells and through the basal lamina..
6. Precapillary sphincters can control the flow of blood into the capillaries.

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SINUSOIDAL CAPILLARIES IN PLACENTA



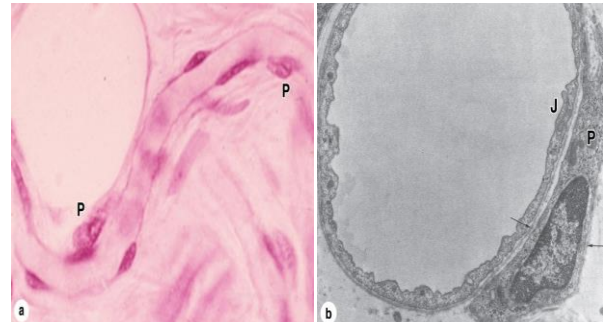
Sinusoidal capillaries or sinusoids generally have much greater diameters than most capillaries and are specialized not only for maximal molecular exchange between blood and surrounding tissue, but also for easy movement of blood cells across the endothelium. The sinusoid (S) shown here is in bone marrow contained in reticular tissue containing adipocytes (A) and masses of hematopoietic cells (H). The endothelium is very thin and cell nuclei are more difficult to find than in smaller capillaries. Ultrastructurally sinusoidal capillaries are seen to have large fenestrae through the cells and large discontinuities between the cells and through the basal lamina.

88

The Pericyte or the perivascular Contractile cell

1. It may completely or incompletely surround a blood capillaries and venules (postcapillary and collecting venules).
2. It is mesenchymal in origin
3. The cell processes usually surround the vessels.
4. The cell plays a contractile function since it contains contractile proteins (myosin and actin).
5. After vessel injury it proliferates and differentiates to form new blood vessels and connective tissue cells.

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A capillary cut transversely, showing the thin wall of one endothelial cell covered by an external basal lamina (arrows). Endothelial cells have numerous transcytotic vesicles and their edges overlap and are bound tightly together with occluding junctions (J). One pericyte (P) is shown, surrounded by its own external lamina. Pericytes can proliferate to form smooth muscle cells when a capillary is transformed into an arteriole or venule after tissue injury and repair.

90

VEINS

- Carry blood towards the heart
- Except for the umbilical and pulmonary veins, veins carry less oxygenated blood.
- They are generally thin-walled than their artery counterparts.
- Their wall is thinner as compared to the diameter of the lumen.
- Structurally classified according to their size as:
 1. Venules
 2. Small veins
 3. Medium-sized veins and
 4. Large veins

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TYPES OF VEINS

1. VENULES

- small tributaries of veins
- have very thin wall, but larger diameter ranging from 10 to 100 μm in diameter.
- are the site in the vasculature where white blood cells leave the circulation to become functional in the interstitial space of surrounding tissues when such tissues are inflamed or infected.
 - Such conditions cause endothelial cells of venules to loosen intercellular junctions and express new protein receptors on their luminal surface.
 - Surface proteins on passing leukocytes bind these receptors, causing the cells to stick to the endothelial cells in a process termed **margination**.
 - This adhesion is quickly followed by emigration from the venule between endothelial cells.
- For deductive purposes, classified as: **post-capillary venules, collecting venules and muscular venules.**

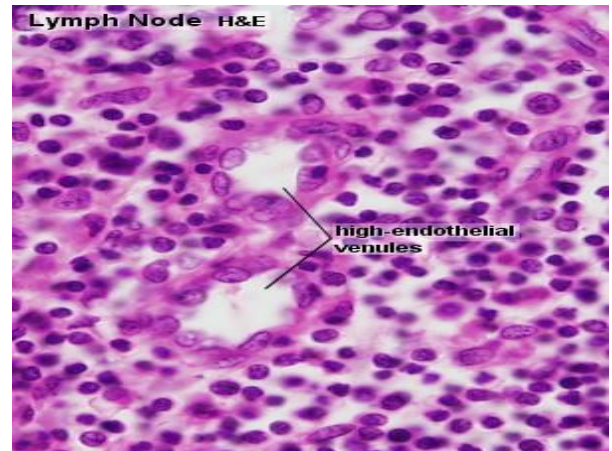
92

Types of Venules

a. Post-capillary venules:

- are the smallest venules
- drain capillary bed/network at the venous end of microvasculature
- Consists of endothelium & discontinuous layer of pericytes
- In some lymphoid organs like the **lymph nodes**, **tonsil**, **peyer's patch** and **appendix**, the endothelium consists of unusual cuboidal cells, and hence called **high endothelial venule or vessel (HEV)**.
- Junctions between endothelial cells of postcapillary venules are the loosest of the microvasculature.
 - At these locations there is a characteristic loss of fluid from the circulatory system during the inflammatory response, leading to edema.

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Post-capillary venules:

- The **endothelium** of postcapillary venules is the principal site of action of vasoactive agents such as histamine and serotonin.
- Response to these agents results in **extravasation of fluid** and emigration of white blood cells from the vessel during inflammation and allergic reactions.
- Postcapillary venules of lymph nodes also participate in the trans mural migration of lymphocytes from the vascular lumen into the lymphatic tissue.
- **Pericytes** form umbrella-like connections with the endothelial cells.

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Types of Venules

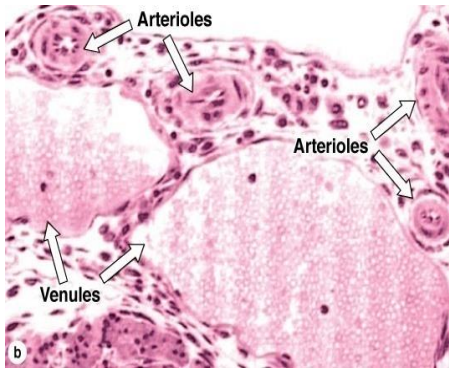
b. Collecting venules:

- Drain several postcapillary venules
- Larger than postcapillary venules, but have similar histology with them except the presence of continuous layer of numerous pericytes around them.

c. Muscular venules

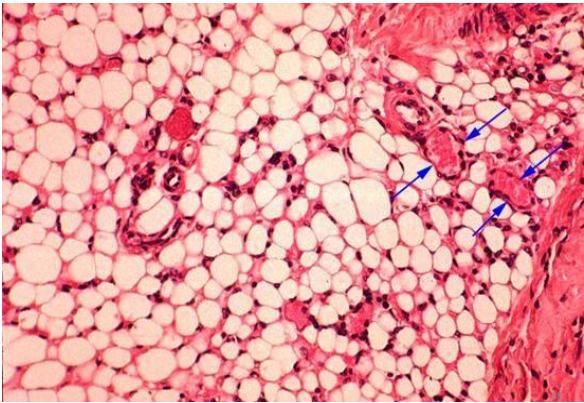
- are the thickest type of the venules
 - have a muscular wall
1. The **intima**: consists of endothelium and its basal lamina
 2. The **media**: contains circumferentially arranged one to three layers of smooth muscle fibers.
 3. The **adventitia**:
 - a more distinct relatively thick tunica adventitia

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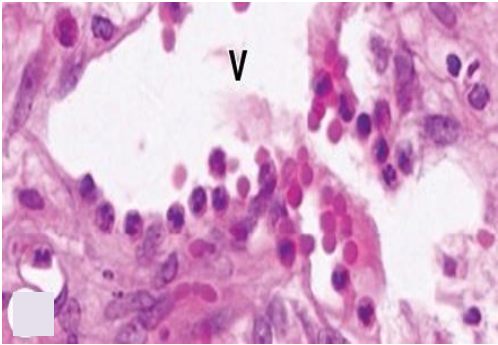
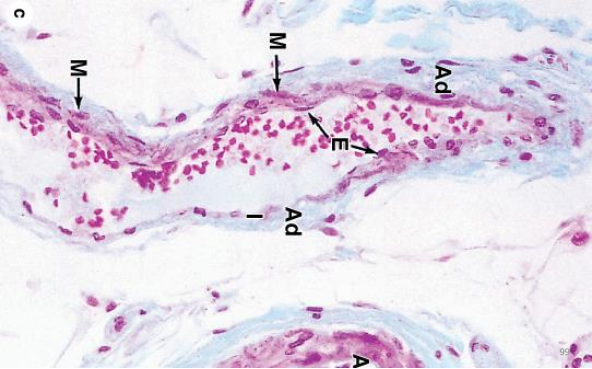


Large collecting venules have much greater diameters than arterioles but the wall is still very thin, consisting of an endothelium with continuous layer of numerous pericytes.

VENULES



Muscular venule: has a better defined tunica media, with as many as three layers of smooth muscle (M) in some areas, a very thin intima (I) of endothelial cells (E), and a more distinct tunica adventitia (Ad). Part of an arteriole (A) is included for comparison.



Venule (V) from an infected small intestine shows several leukocytes adhering (margination) to and migrating across the endothelium.

Types of Veins----- con't

2. SMALL-SIZED VEINS (0.1 – 1 mm)

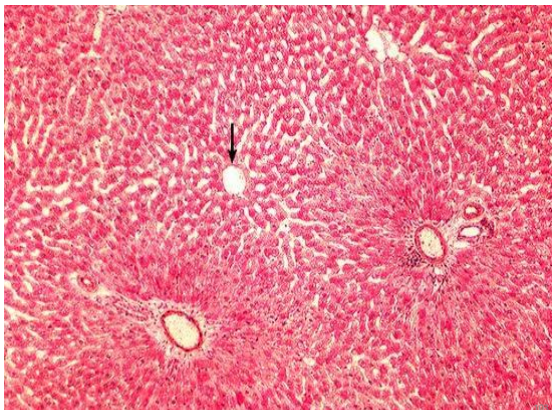
- E.g. central vein of liver lobule, trabecular vein of spongy bone
- have a diameter of 0.1 mm to 1 mm.
- The intima is thin with a thin subendothelial layer.
- The media contains interwoven smooth muscles.
- The adventitia is rich with collagen fibers.
- Contain semilunar shaped valves in their intima.

101

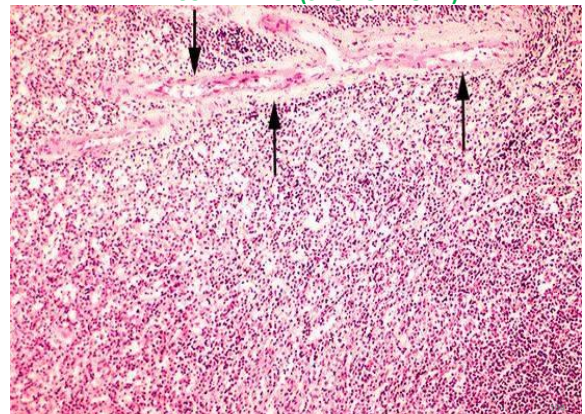
Small-sized Veins

- Most of the veins in the body are **small veins**, with diameters less than one centimeter.
- are usually located in parallel with corresponding small-sized arteries.
- The intima is thin usually with a thin subendothelial layer with out internal elastic lamina and the media consists of small bundles of smooth muscle cells intermixed with reticular fibers and a delicate network of elastic fibers.
- The collagenous adventitial layer is well-developed

CENTRAL VEIN



TRABICULAR VEIN (SPONGY BONE)

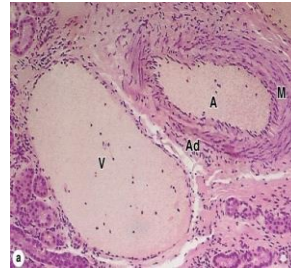


3. MEDIUM-SIZED VEINS: 1 to 10 mm

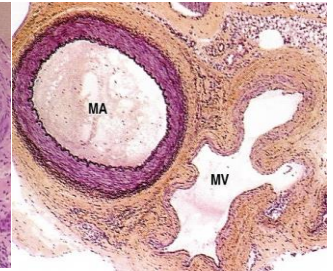
e.g. Femoral vein, cephalic vein

- constitute the majority of the named veins, usually located in parallel with corresponding muscular arteries.
- The intima is thin with no internal elastic lamina
- The media is formed of thin layer, which contains small bundles of smooth muscle fibers.
 - Some elastic and reticular fibers are included in the wall of the tunica media.
- Tunica adventitia is well developed in the wall of medium sized veins, it contains bundles of collagenous fibers.
- Valves are common structure in the lumen of veins.
- They are intimal luminal foldings.

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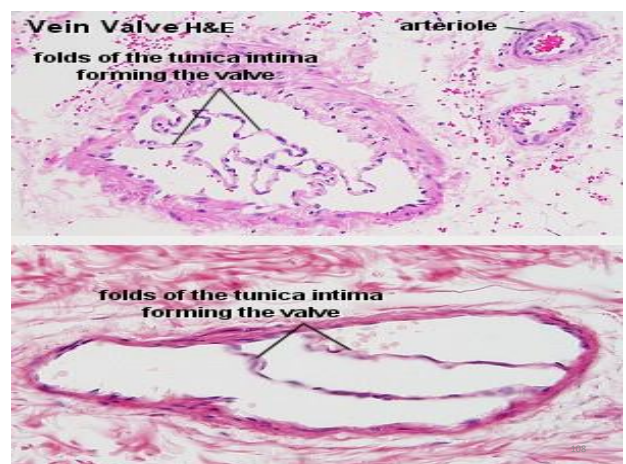
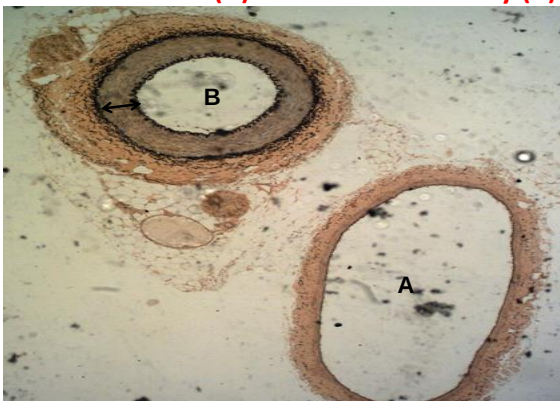


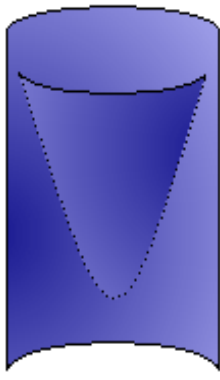
Micrograph of small vein (V) shows a relatively large lumen compared to the small muscular artery (A) with its thick media (M) and adventitia (Ad).



Micrograph of a medium-sized vein (MV) showing a thicker wall, but still less prominent than that of the accompanying muscular artery (MA). Both the media and adventitia are better developed, but the wall is often folded around the relatively large lumen. 106

Medium-sized Vein (A) and Muscular artery (B)





pocket valve



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4. LARGE VEINS (>10 mm)

- These includes the main trunks of the heart (SVC, IVC and hepatic portal vein)
- have well developed intima.
- The tunica media is much thinner with few layers of smooth muscles.
- The adventitia is by far the thickest part of the vascular wall.

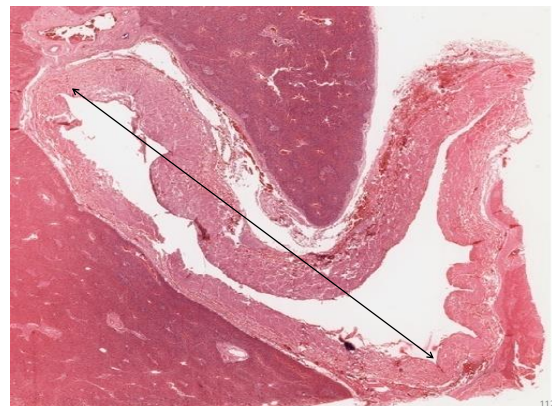
110

Structure of Large veins (e.g. portal vein, Inferior vena cavae)

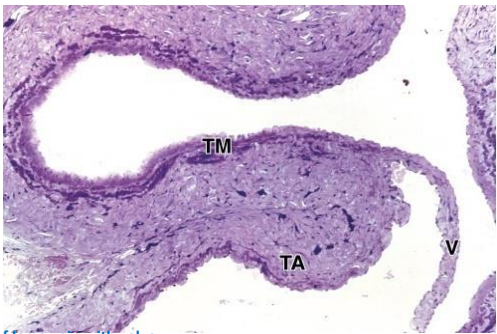
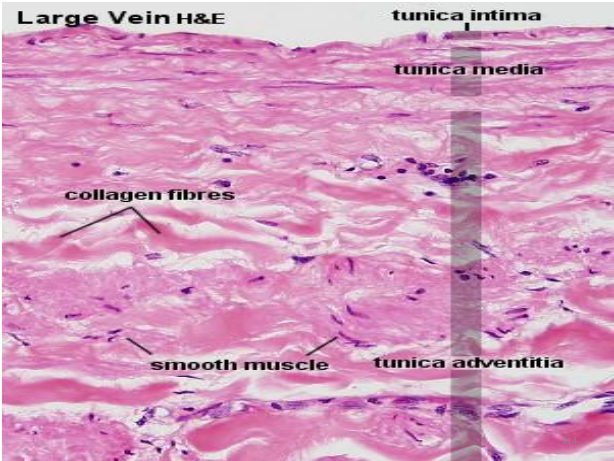
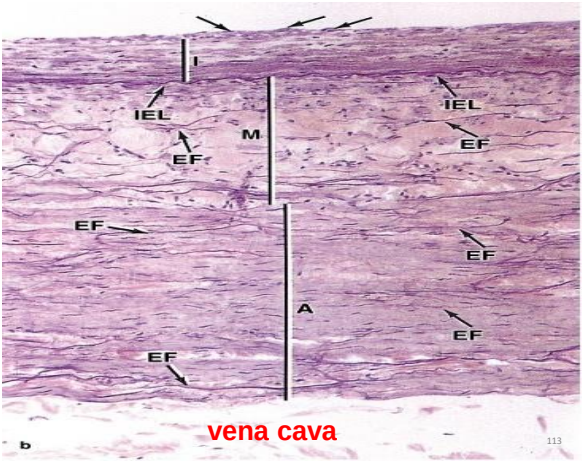
1. Tunica intima is well developed
2. Tunica media is thin and may contain few layers of concentrically arranged smooth muscle fibers
3. Tunica adventitia is the thickest layer, it contains longitudinal bundles of smooth muscle fibers, collagen fibers and elastic fibers.

111

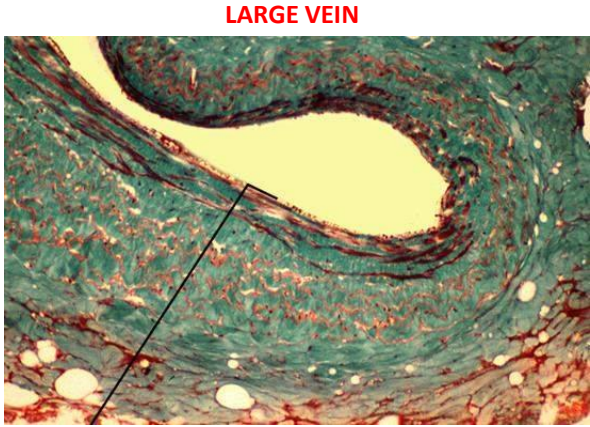
PORTAL VEIN



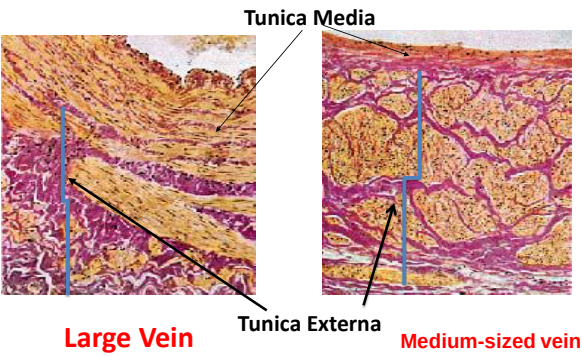
112



Wall of large vein with valve.
Large veins have a muscular tunica media (TM) that is very thin compared to the tunica adventitia (TA) which is composed of dense irregular connective tissue and muscle. The wall is often folded as shown here. The tunica intima here projects into the lumen as a valve (V), composed of the subendothelial connective tissue with endothelium on both sides.

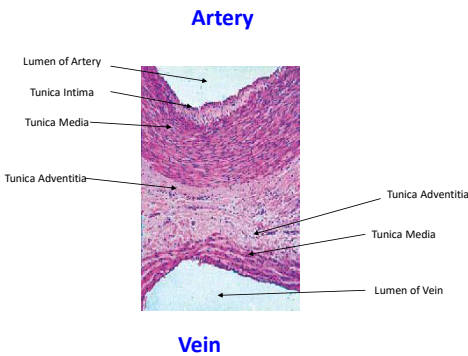


Veins



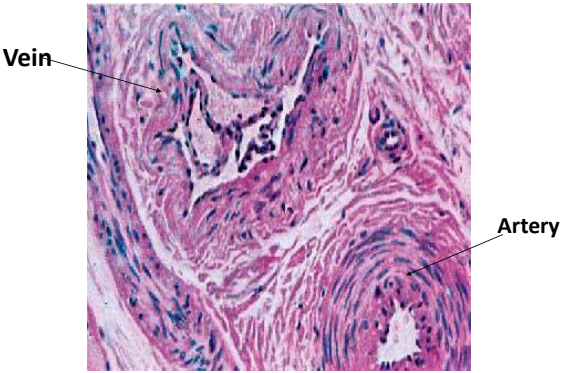
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Amalgamated Artery and Vein



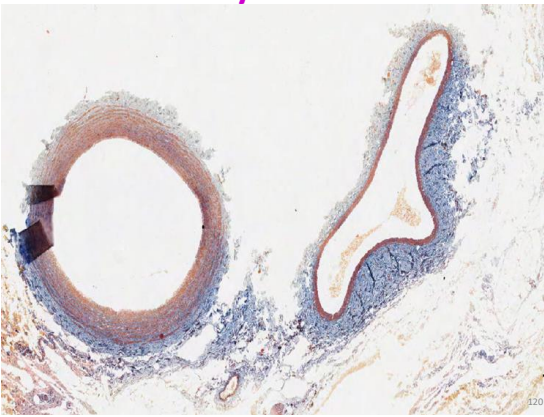
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Artery and Vein



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Artery and Vein



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Artery, Vein, a lymphatic vessel and small nerves



Parts of the Lymphatic Vascular System

1. Lymphatic capillaries

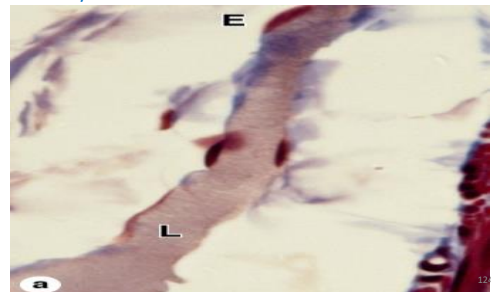
- The lymphatic system originates from lymphatic capillaries which are blind ended vessels lined by very thin endothelium, having **no fenestrae**, **no fascia occludens** and **interrupted/discontinuous basal lamina**.
- 10-50 μm in diameter
- do not contain fenestrations but have very thin and porous basal lamina.
- **Lacteals**, are large lymph capillaries in the core of intestinal villi for absorption of lipids
- They all drain into lymphatic efferent vessels and then ultimately to two large trunks, thoracic duct and right lymphatic duct.

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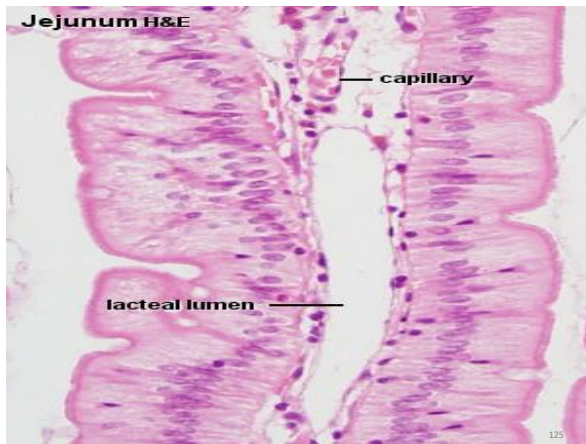
- The lymphatic capillaries are kept open by **anchoring filaments** consisting of a core of **elastin** covered by **flaps of endothelium**.
- Lymphatic capillaries drain interstitial, extravassated fluids produced when the plasma forced from the microvasculature by hydrostatic pressure does not all return to blood by the action of osmotic pressure.
- These capillaries absorb tissue fluid and transfer it to larger lymphatic vessels until they reach the thoracic duct (the largest duct in human body), and the right lymphatic duct which drain into the blood stream (venous side).

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Lymphatic capillaries : Micrograph showing a lymphatic capillary filled with this fluid called **lymph** (L). Lymphatics are blind-ended vessels with a wall of very thin endothelial cells (E) and are quite variable in diameter (10-50 μm). Lymph is rich in proteins and other material and often stains somewhat better than the surrounding ground substance, as seen here. X200. Mallory trichrome



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Parts of the Lymphatic Vascular System

2. Lymphatic vessels

- similar to veins but have thinner walls and have no clear cut divisions of the layers.
- contain numerous internal valves which prevents backflow.
- Lymphatic vessels are found in almost all tissues except the CNS, cartilage, bone, bone marrow, thymus cortex, placenta, cornea and teeth.

3. Lymphatic ducts

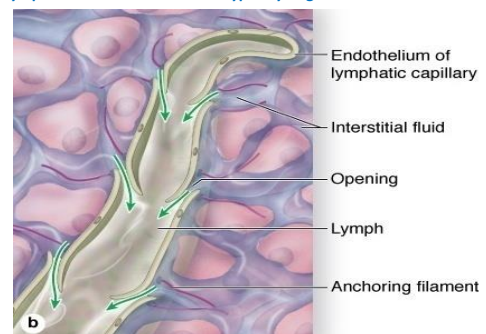
- similar structure to veins.
- contain longitudinal fibers.
- adventitia is underdeveloped.
- Contain vasa vasorum.

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- Besides gathering interstitial fluid as lymph and returning it to the blood, the lymphatic system of vessels is a major distributor of lymphocytes, antibodies, and other immune components which it picks up at lymph nodes and other lymphoid tissues.
- The larger lymphatics have a structure similar to that of veins except that they have thinner walls and lack a clear-cut separation between the tunics. They also have more numerous internal valves.
- The lymphatic vessels are often dilated and assume a nodular, or beaded, appearance between the valves.
- As in veins, lymphatic circulation is aided by external forces (e.g., contraction of surrounding skeletal muscle) and unidirectional lymph flow is mainly a result of the many valves.
- Contraction of smooth muscle in the walls of larger lymphatic vessels also helps to propel lymph toward the heart.

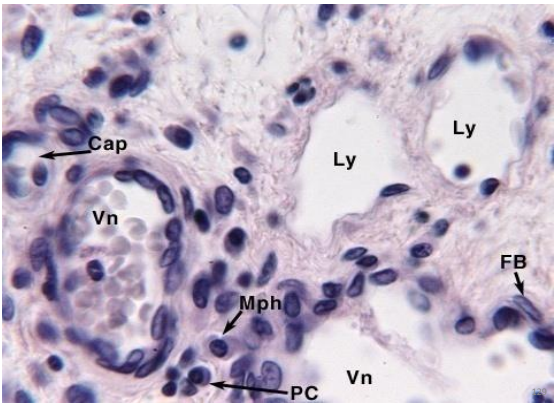
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Diagram indicating details of lymphatics, including the openings between the endothelial cells. The openings are held in place by anchoring filaments containing elastin and are covered by flaps of endothelium. Interstitial fluid enters primarily via these openings and the endothelial folds prevent backflow of lymph into tissue spaces. Lymphatic endothelial cells are typically larger than those of blood vessels.

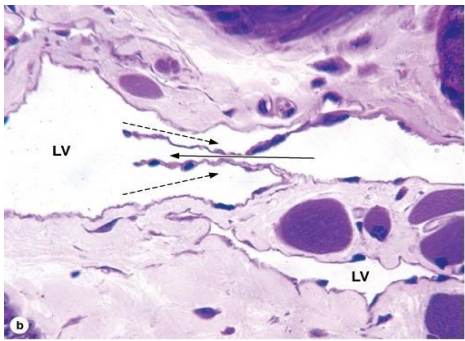
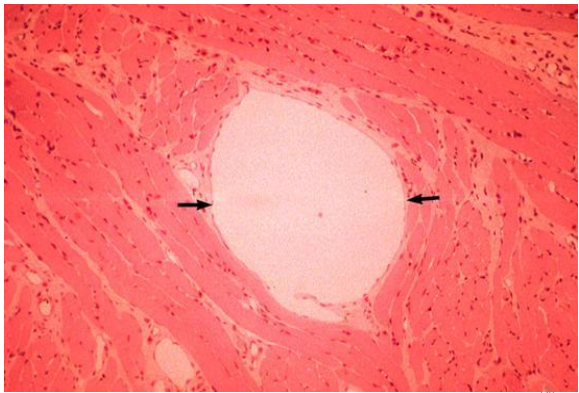


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LYMPHATIC VESSELS



LYMPHATIC VESSEL



Lymphatic vessel (LV) in muscle is cut longitudinally showing a valve, the structure responsible for the unidirectional flow of lymph. The solid arrow shows the direction of the lymph flow, and the dotted arrows show how the valves prevent lymph backflow. The lower small lymphatic vessel is a lymphatic capillary with a wall consisting only of endothelium.

Valves of Lymphatic vessel

